

AWS CloudWatch Lab: Collect EC2 Application Logs Using CloudWatch Agent

Summary:-

In this lab, an **Amazon EC2** instance hosting a sample web application was configured to send application logs to **Amazon CloudWatch Logs** using the **CloudWatch Agent**. An **IAM Role** with the required permissions was attached to the EC2 instance, the CloudWatch Agent was installed and configured, and a **CloudWatch Log Group** was created to centrally store application logs. Finally, the application generated log events that were successfully monitored in real time through CloudWatch Logs.

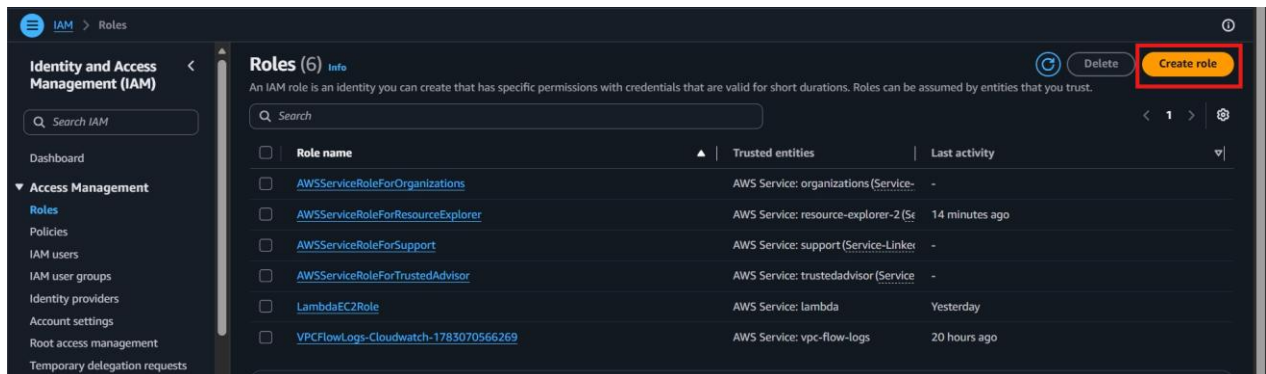
Let's Start:-

- **Amazon CloudWatch Agent** enables Amazon EC2 instances to send application logs and system metrics to **Amazon CloudWatch**.
- Application logs are collected from a specified log file and stored centrally in **CloudWatch Log Groups**.
- An **IAM Role** with the required permissions must be attached to the EC2 instance before installing the CloudWatch Agent.

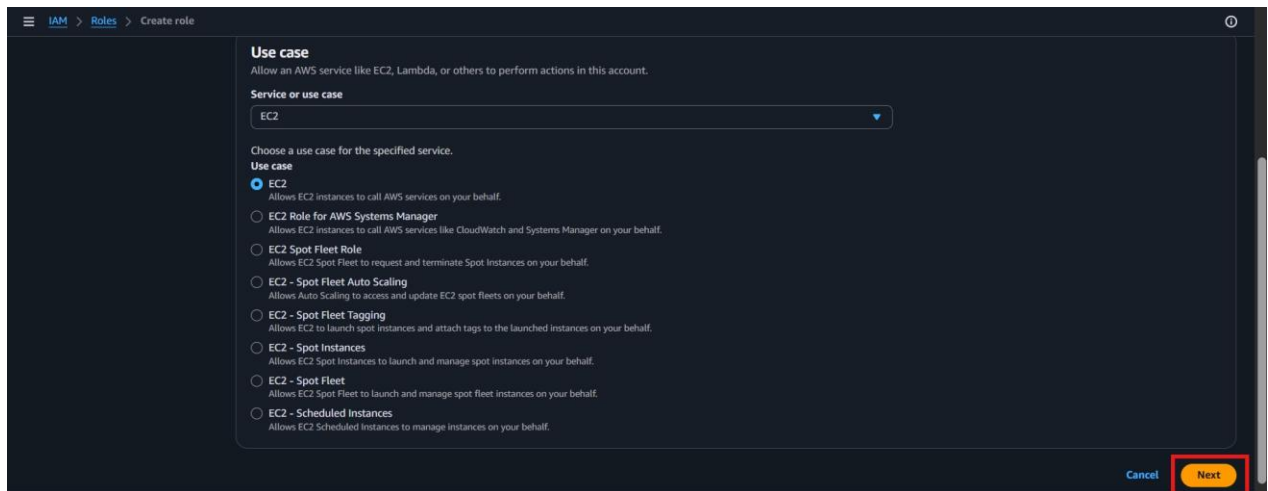
Lab Steps:-

Step 1 – Create an IAM Role

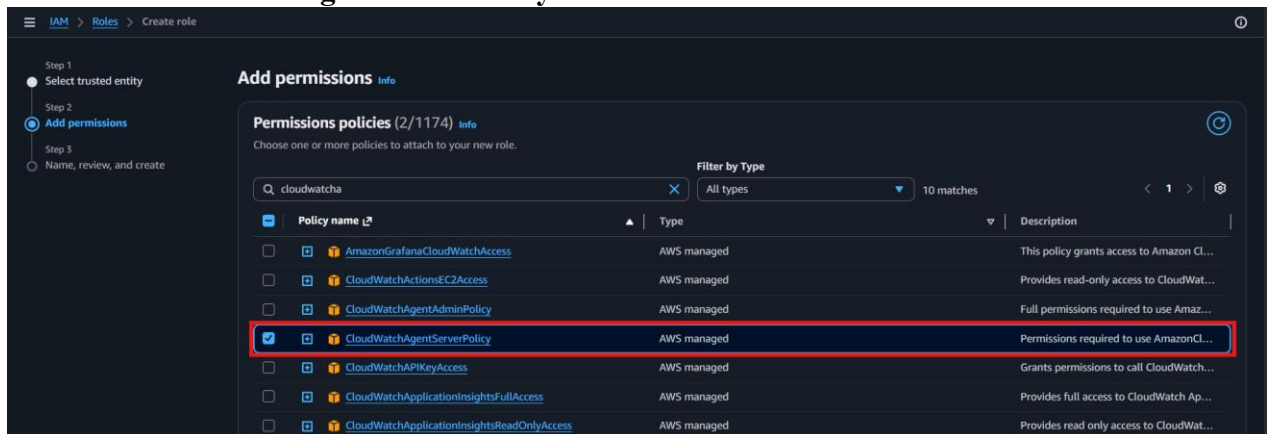
- Sign in to the **AWS Management Console**.
- Navigate to **IAM**.
- Create a new IAM Role for **Amazon EC2**.



- Under Trusted Entity Type, select **AWS Service**.
- Under Use Case, select **EC2**.
- Click **Next**.



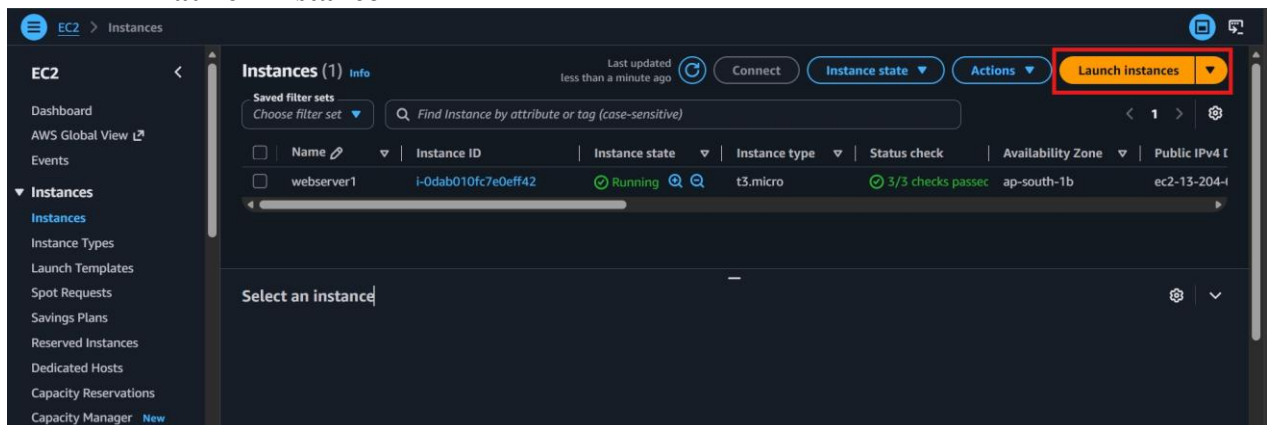
- Attach the following managed policies:
 - **AmazonSSMManagedInstanceCore**
 - **CloudWatchAgentServerPolicy**



- Save the role (e.g., **LogAccessRole**).

Step 2 – Launch an EC2 Instance

- Navigate to **Amazon EC2**.
- Click **Launch Instance**.



- Enter an instance name (e.g., **Application Server**).

Name and tags info

Name
Application-Server [Add additional tags](#)

Application and OS Images (Amazon Machine Image) info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-0d351f1b76da30161 (64-bit (x86), uefi-preferred) / ami-0a2f5ec7b370b274 (64-bit (Arm), uefi)

Summary

Number of instances info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...[read more](#)
ami-0d351f1b76da30161

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

- Select **Amazon Linux 2023**.
- Choose a **Free Tier Eligible** instance type.
- Select an existing **Key Pair**.
- Use the **Default VPC**.
- Enable **Allow HTTP Traffic (Port 80)**.

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required
Testkey [Create new key pair](#)

Network settings info [Edit](#)

Network info
vpc-088c2252c9d19ada2

Subnet info
No preference (Default subnet in any availability zone)

Auto-assign public IP info
Enable

Firewall (security groups) info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

[Create security group](#) [Select existing security group](#)

We'll create a new security group called 'launch-wizard-10' with the following rules:

☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

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[Cancel](#) [Launch instance](#) [Preview code](#)

Step 3 – Attach the IAM Role

- Expand **Advanced Details**.
- Under **IAM Instance Profile**, select the previously created **LogAccessRole**.

Advanced details info

Domain join directory info
Select [Create new directory](#)

IAM instance profile info
Log-Access-Role
arn:aws:iam::386361158340:instance-profile/Log-Access-Role [Create new IAM profile](#)

Hostname type info
IP name

DNS Hostname info
☒ Enable IP name IPv4 (A record) DNS requests
☒ Enable resource-based IPv4 (A record) DNS requests
☒ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery info
Select

Summary

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Amazon Linux 2023 AMI 2023.12...[read more](#)
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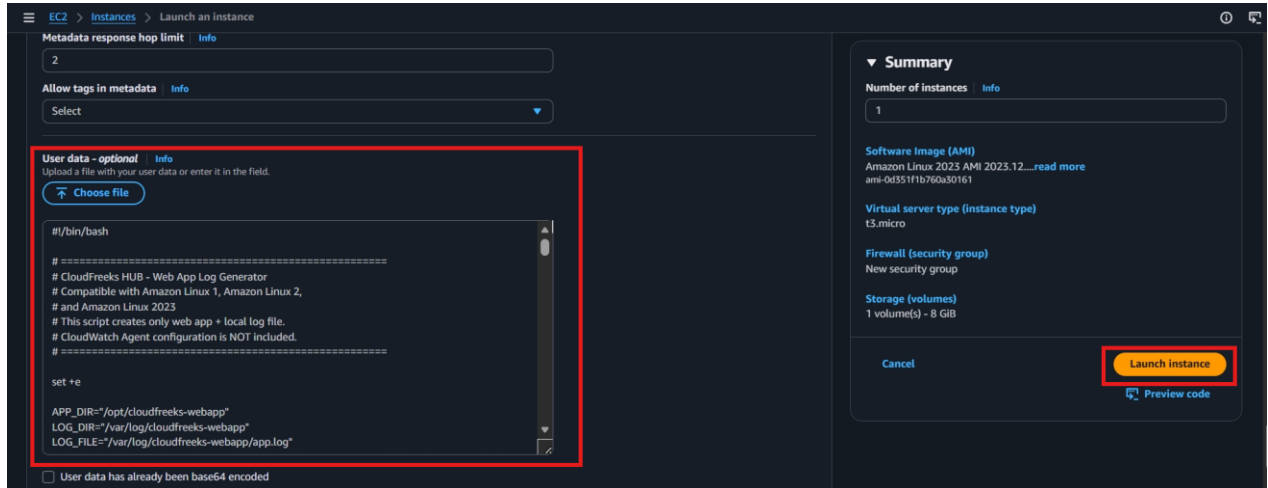
Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Step 4 – Add the User Data Script

- Open the provided **User Data Script (.sh)** from the lab resources.
- Copy the complete script.
- Paste it into the **User Data** section.

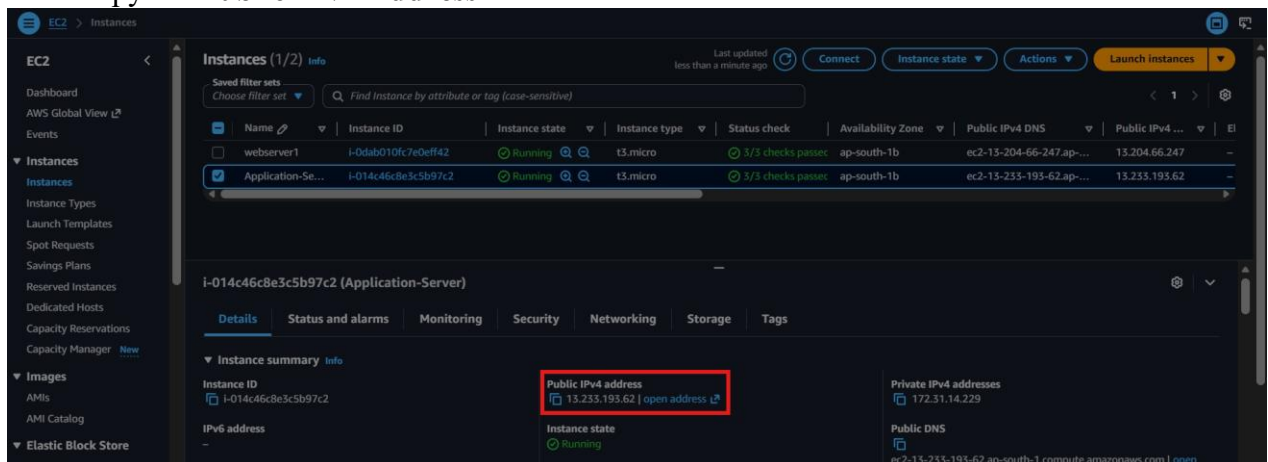


- Click **Launch Instance**.
- Wait until the EC2 instance reaches the **Running** state.

Step 5 – Verify the Web Application

Step 5 – Connect to the EC2 Instance Using SSH

- Copy the **Public IPv4 Address** of the EC2 instance.

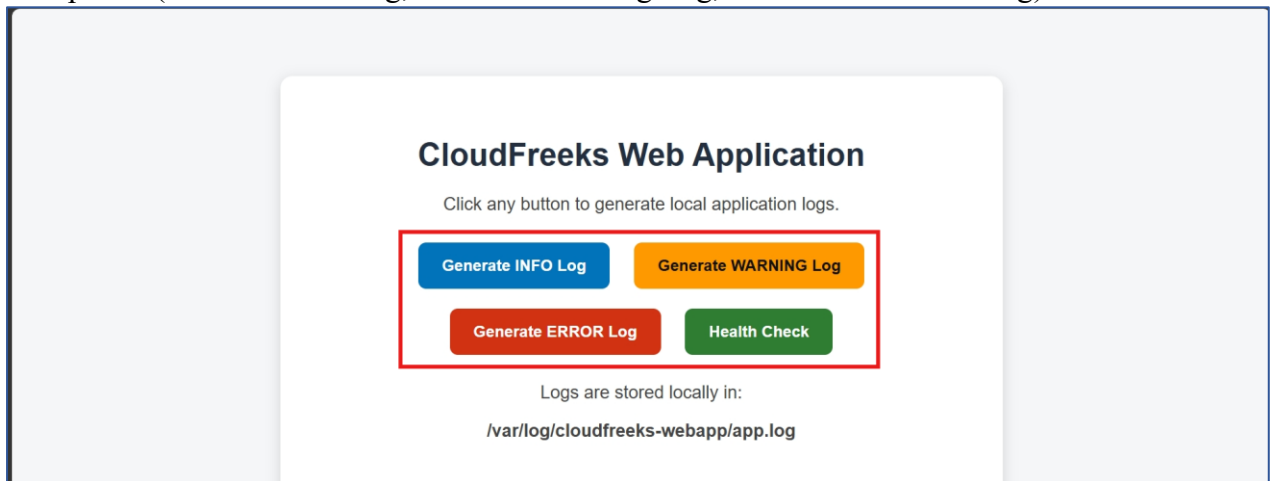


- Open Command Prompt or Terminal on your local machine.
- **Enter:**
`ssh -i Testkey.pem ec2-user@<Public IPv4 Address>`
- Press **Enter**.

- It will **show**:
The authenticity of host '<Public IPv4 Address>' can't be established. Are you sure you want to continue connecting (yes/no)?
- Type **yes** and press **Enter**.
 - Wait for the **SSH connection to be established**.
 - Verify that the *Amazon Linux 2023* login banner and the **Last login** message are displayed, indicating that the connection to the EC2 instance was successful.

Step 6 – Verify the Web Application

- Copy the Public IPv4 Address of the EC2 instance.
- Open a **web browser**.
- Enter:
http://<Public IPv4 Address>
- Verify that the web application loads successfully and displays the log generation options (Generate Info Log, Generate Warning Log, and Generate Error Log).



Step 7 – Verify Local Application Logs

- Connect to the EC2 instance using **SSH**.
- Navigate to the application log location.
- **Run** the following command in terminal to monitor the log file:
tail -f /var/log/cloudfrees-webapp/app.log
- Generate **Info**, **Warning**, and **Error** logs from the web application.

CloudFreeeks Web Application

Click any button to generate local application logs.

Generate INFO Log

Generate WARNING Log

Generate ERROR Log

Health Check

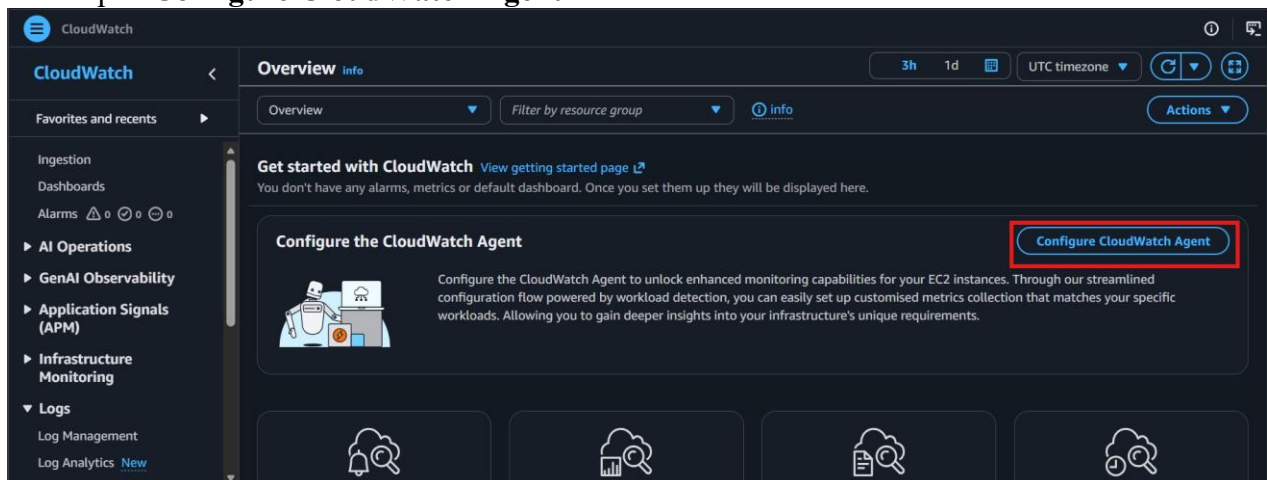
Logs are stored locally in:

`/var/log/cloudfreeeks-webapp/app.log`

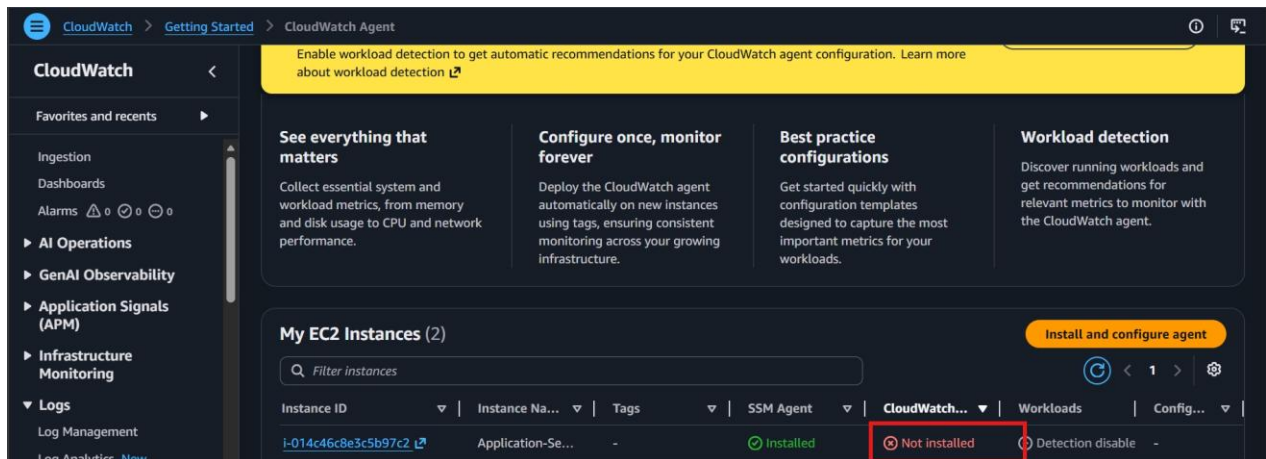
- Verify that new log entries appear in the **terminal**.

Step 8 – Open CloudWatch Agent Configuration

- Navigate to **Amazon CloudWatch**.
- Open **Configure CloudWatch Agent**.



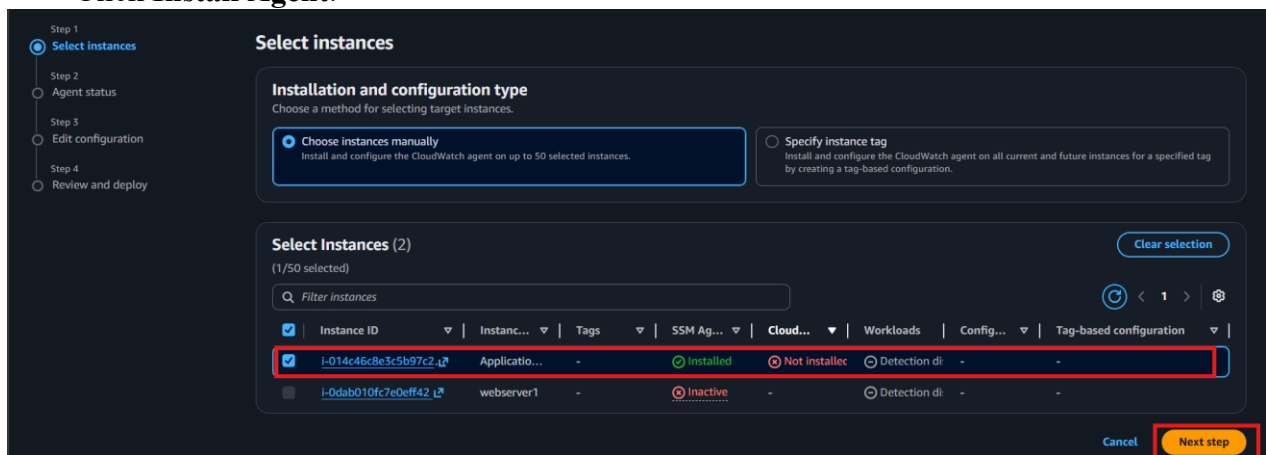
- Verify that the EC2 instance appears in the list.
- Confirm that the CloudWatch Agent is **not installed**.



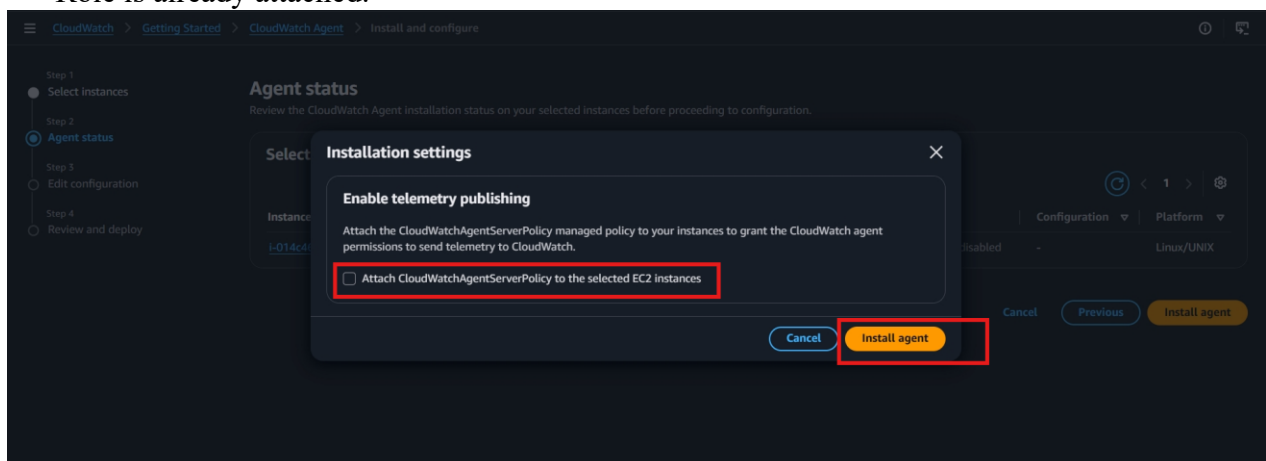
- Click **Install and Configure Agent**.

Step 9 – Install the CloudWatch Agent

- Select the EC2 instance.
- Click **Next**.
- Click **Install Agent**.



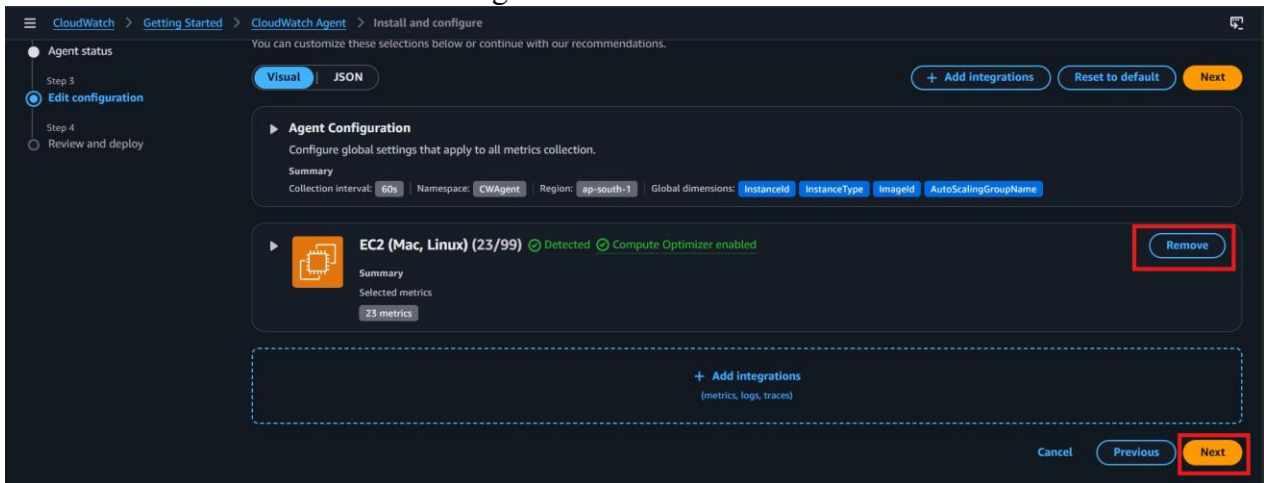
- If prompted, remove any additional IAM permission request since the required IAM Role is already attached.



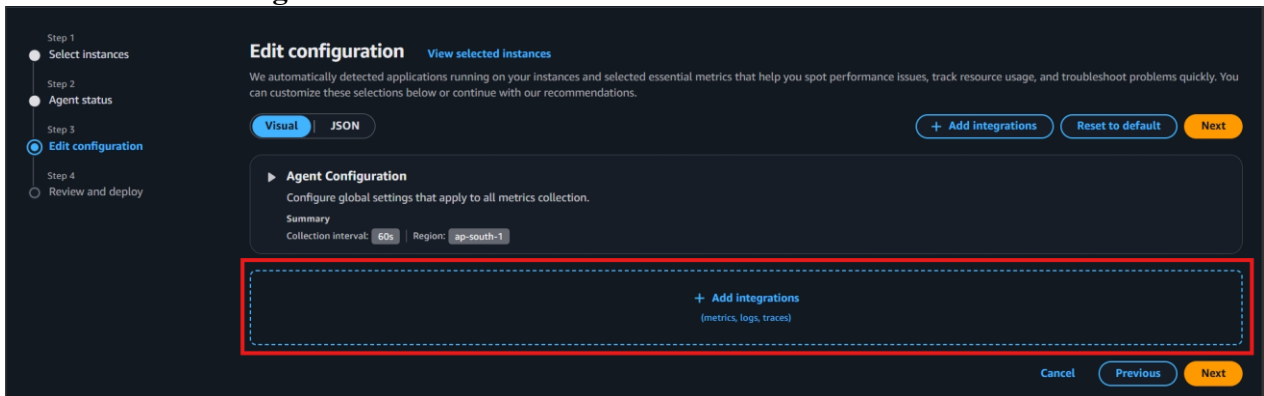
- Wait until the CloudWatch Agent installation completes.

Step 10 – Configure Log Collection

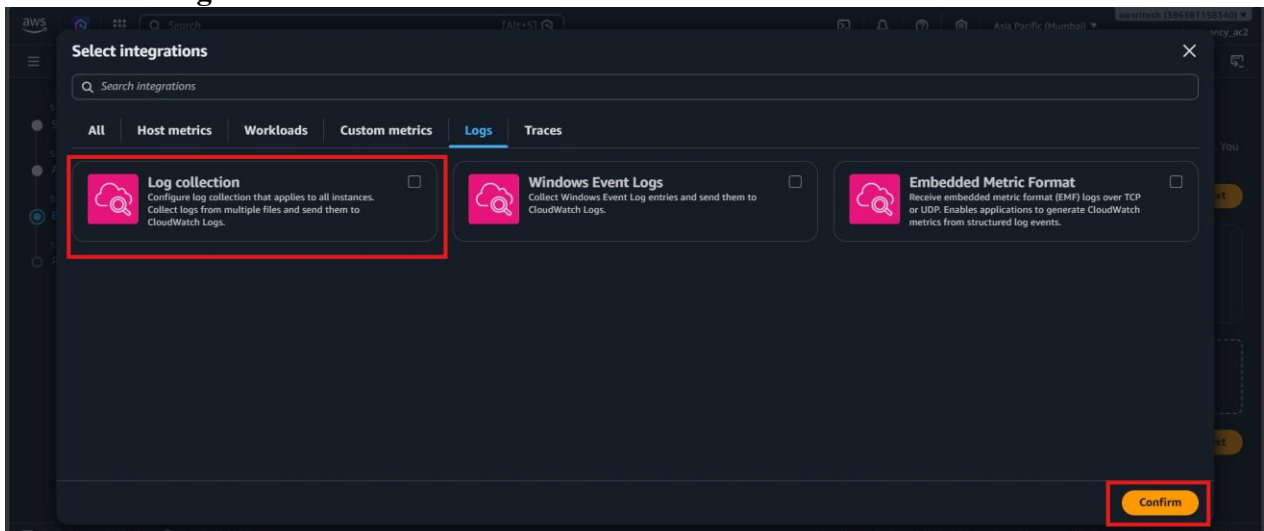
- Remove the default **Metrics** integration.



- Click **Add Integration**.



- Select **Logs**.



- Choose **Simple Log Collection (Linux)**.

- Click **Confirm**.

Step 11 – Configure the Log File

- Enter the application log file path:
/var/log/cloudfreeks-webapp/app.log
- Enter a **Log Group** name (e.g., **MyApplicationLogGroup**).

The screenshot shows the 'Install and configure' page for the CloudWatch Agent. The 'File path' field is set to '/var/log/cloudfreeks-webapp/app.log'. The 'Log group name' field is set to 'MyApplicationLogGroup'. The 'Log stream name' field is set to '(instance_id)'. The 'Include filter' field is set to 'e.g., ERROR|WARN'. The 'Exclude filter' field is set to 'e.g., DEBUG'. The 'Timestamp format' field is set to '%Y-%m-%dT%H:%M:%S'.

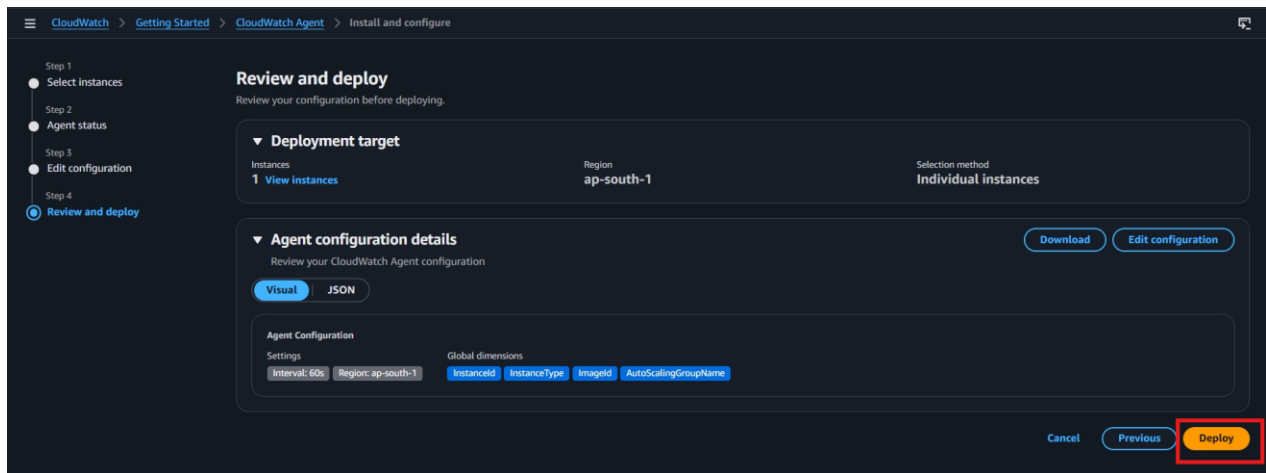
- Keep the default **Log Stream** name.
- Set the **Retention Period** to **3 Days**.

The screenshot shows the 'Install and configure' page for the CloudWatch Agent. The 'Timestamp format' field is set to '%Y-%m-%dT%H:%M:%S'. The 'Multi-line start pattern' field is set to '(timestamp_format)'. The 'Retention' dropdown is set to '3 days'. The 'Enable auto removal' checkbox is unchecked. The 'Add log file' button is visible. The 'Add integrations' button is also visible. The 'Next' button is highlighted.

- Click **Next**.

Step 12 – Deploy the Configuration

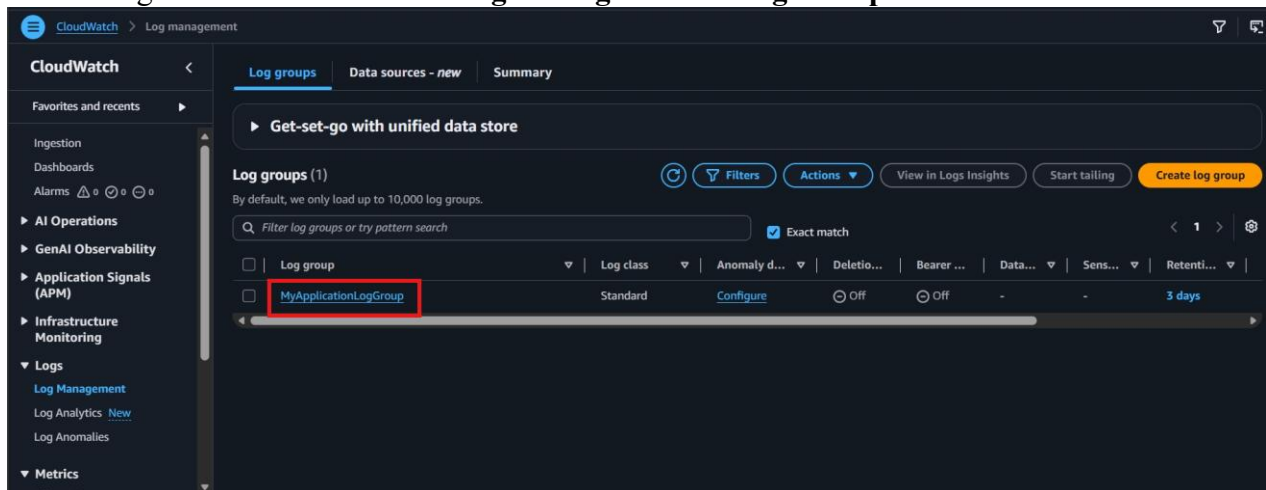
- Review the CloudWatch Agent configuration.
- Click **Deploy**.



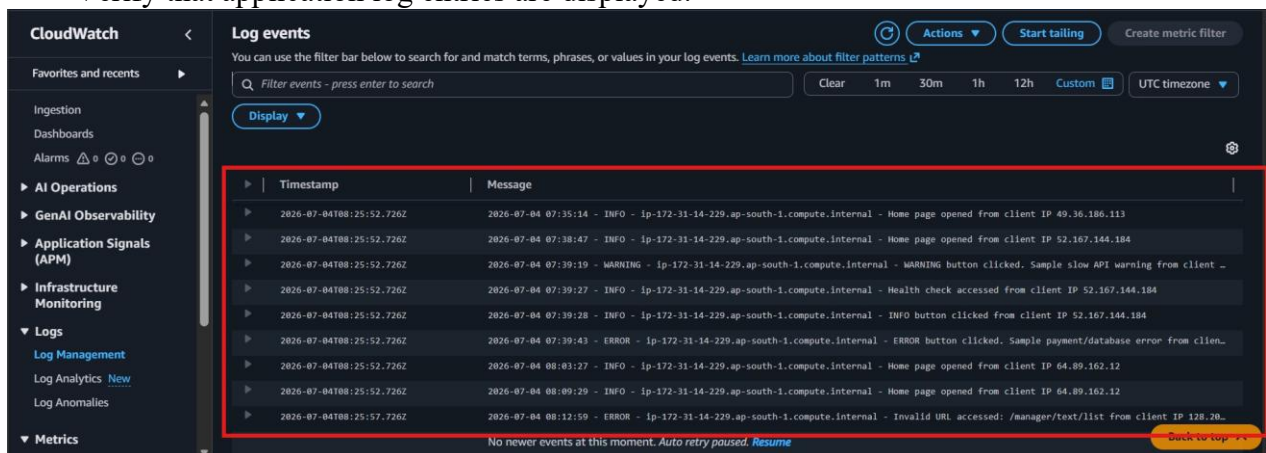
- Wait until the deployment completes successfully.

Step 13 – Verify the Log Group

- Navigate to **CloudWatch** → **Log Management** → **Log Groups**.

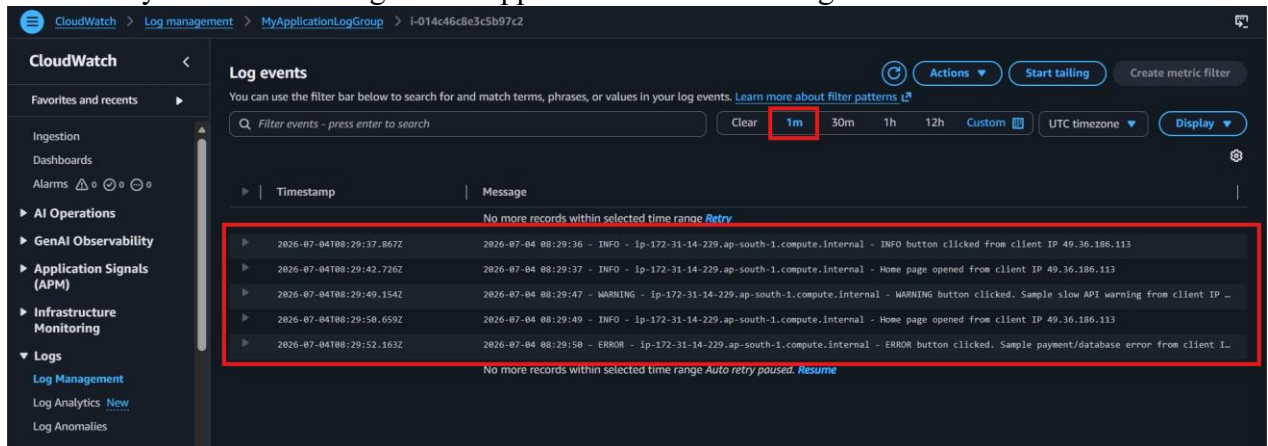


- Open the newly created **Log Group**.
- Open the generated **Log Stream**.
- Verify that application log entries are displayed.



Step 14 – Generate New Application Logs

- Return to the web application. Select 1minute in duration bar.
- Generate **Info**, **Warning**, and **Error** log messages.
- Refresh the CloudWatch Log Stream.
- Verify that the latest log entries appear in CloudWatch Logs.



Step 15 – Monitor Live Logs in Terminal

- Click **Start Tailing** within the CloudWatch Log Stream.
- Generate additional log messages from the web application.
- Verify that the new log events are displayed in real time in the terminal.

Verification

- An IAM Role with the required CloudWatch Agent permissions was created successfully.
- An EC2 instance hosting the sample web application was launched.
- The CloudWatch Agent was installed and configured successfully.
- Application logs were collected from the specified log file.
- A CloudWatch Log Group and Log Stream were created automatically.
- Newly generated application logs appeared successfully in **Amazon CloudWatch Logs**.
- Real-time log streaming using **Start Tailing** was verified successfully.